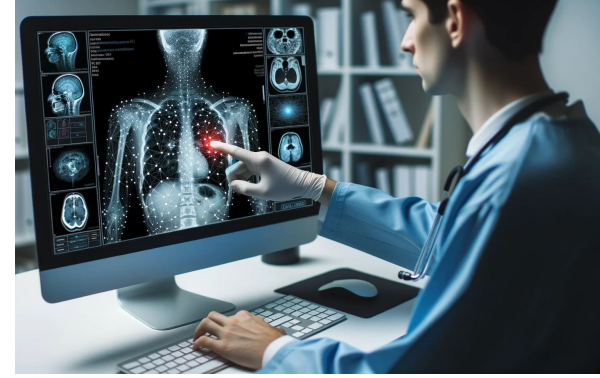
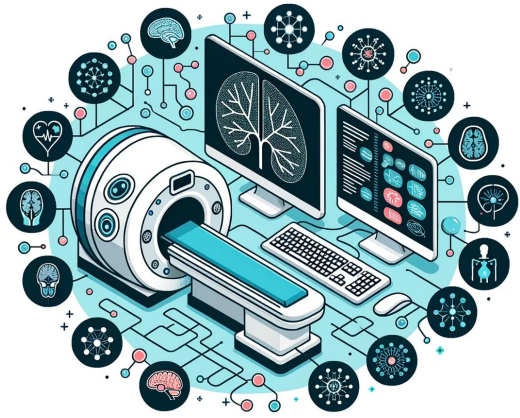




# Session 1

## Introduction to Generative AI in Healthcare

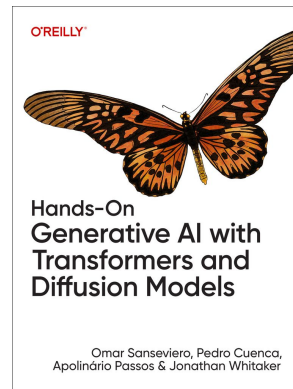
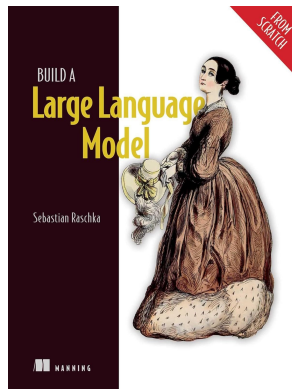
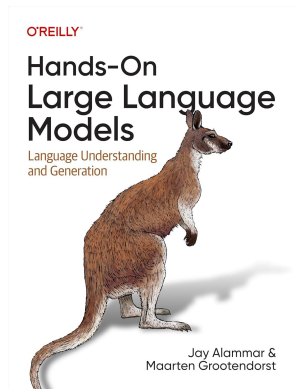
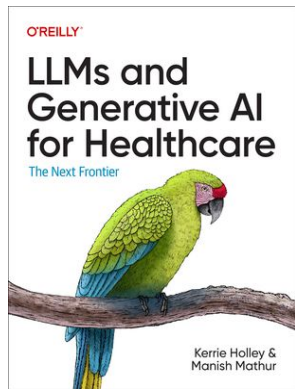
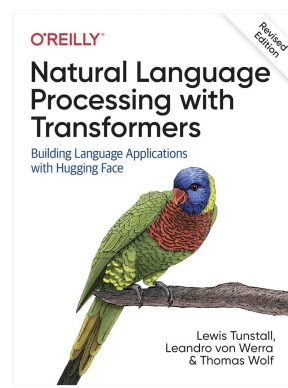
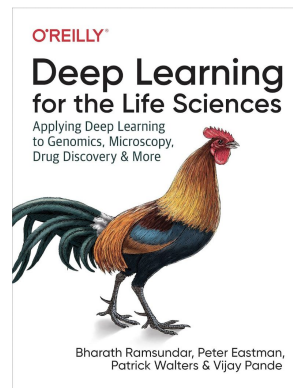
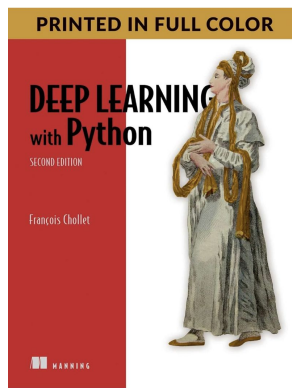
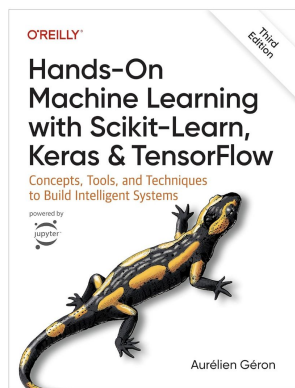
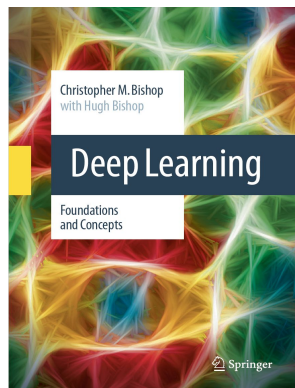
Sep 06 2025

# Course structure and objectives

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- **Format** Lecture + Interactive Lab sessions (hands-on coding in Jupyter Notebooks)
  - **Resources** All slides, notebooks, and readings on Canvas; code repositories on **GitHub**
  - **Evaluation** Homework assignments, Capstone project (individual)
  - **Learning Outcomes**
    - Design, train, and debug deep-learning models for medical imaging, drug discovery, and related tasks.
    - Apply state-of-the-art generative models (e.g., GANs, VAEs, diffusion) to healthcare data.
    - Assess model safety, bias, and regulatory compliance; communicate findings to technical and clinical stakeholders.
  - **Prerequisites** Proficiency in **Python** programming, Working knowledge of **PyTorch**, Core machine-learning concepts (supervised / unsupervised, evaluation metrics)
-

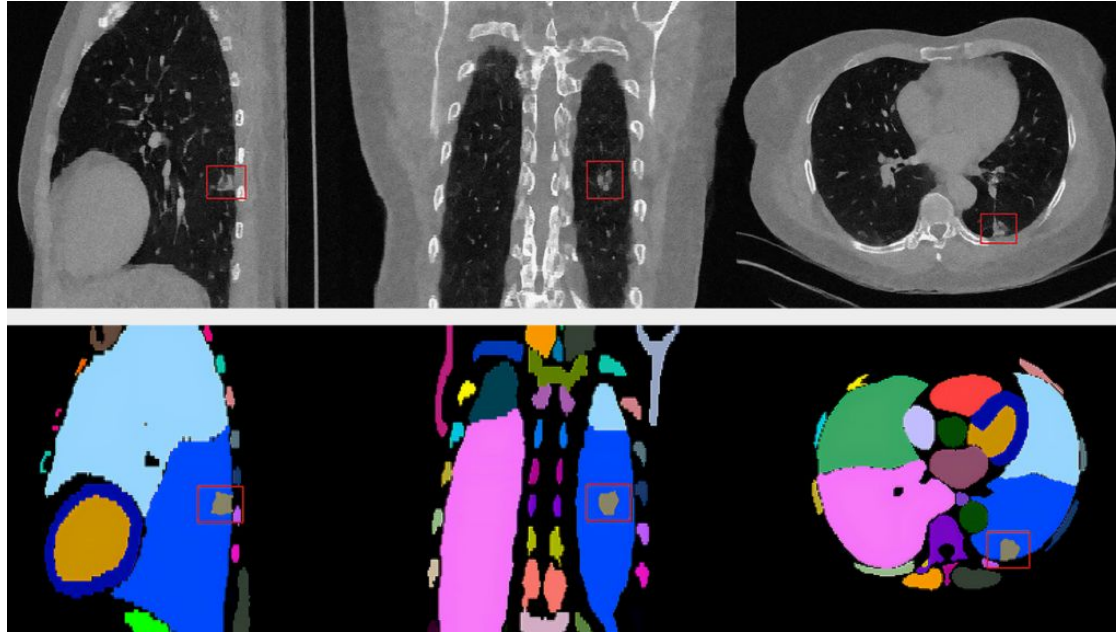
# Some useful resources



# AI in Medical Imaging

---

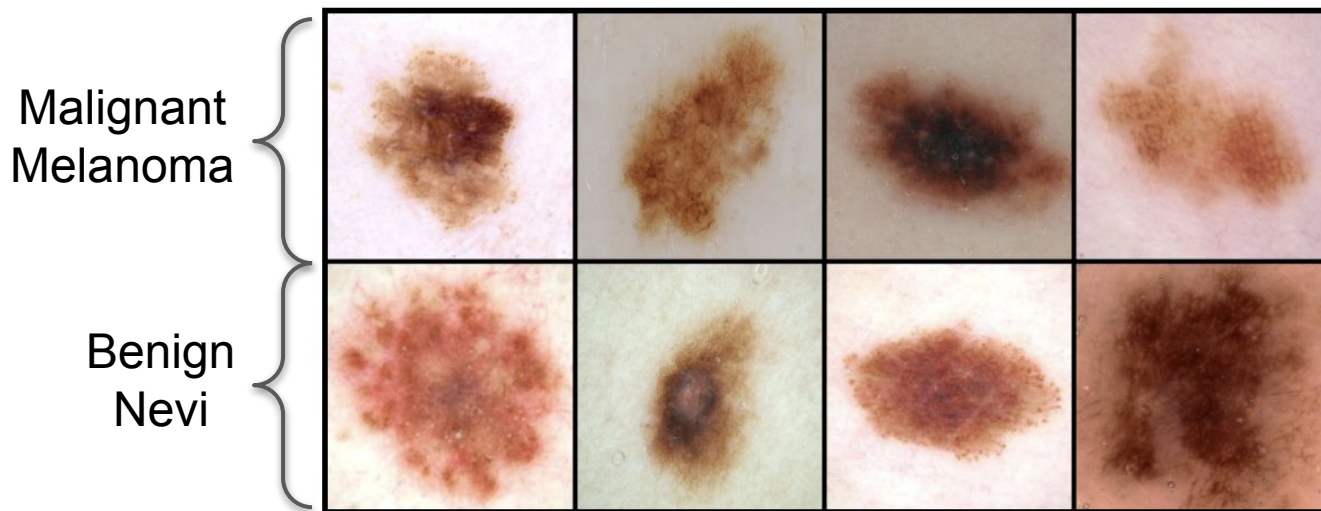
**Data Availability:** Electronic Health Records (EHRs), medical imaging repositories, genomics data.



# AI for Medical Diagnosis

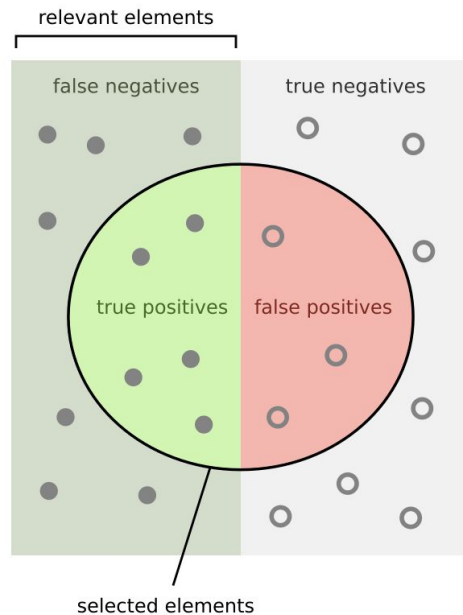
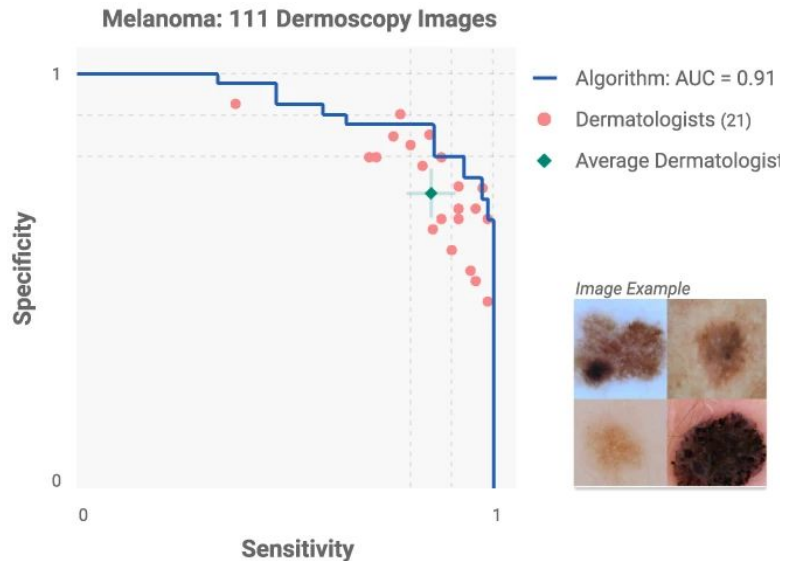
---

**Growing Need:** Address physician shortages, reduce medical errors, accelerate drug discovery, personalized medicine.



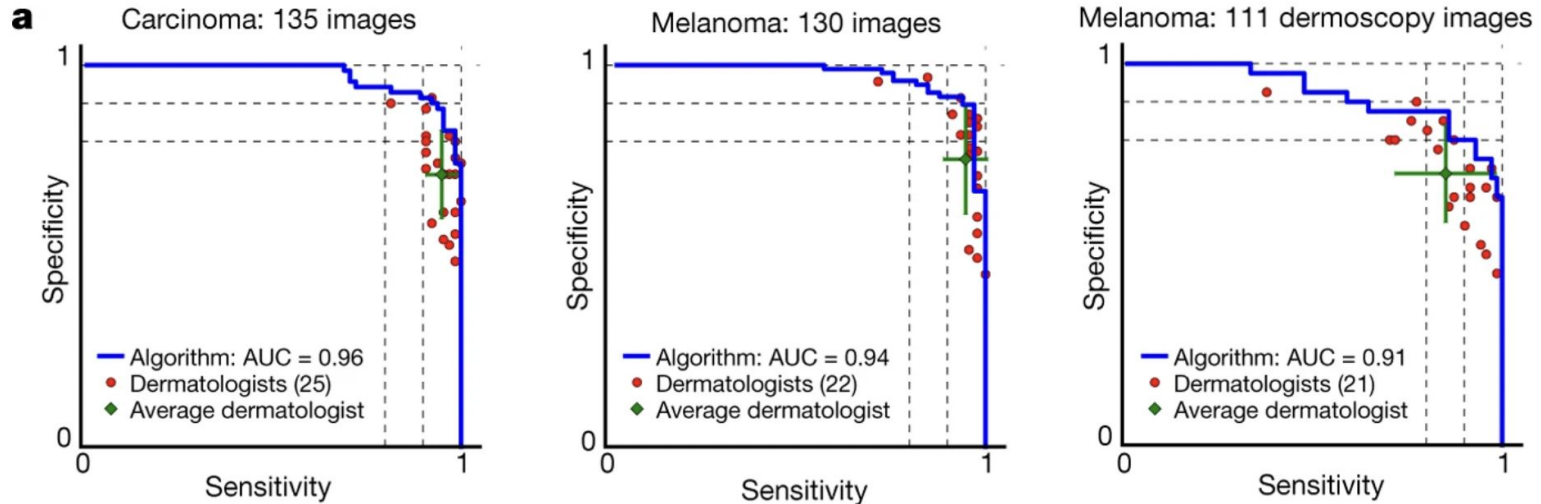
# AI in Medical Diagnosis

**Transformation Potential:** From diagnostics to drug discovery, deep learning is driving innovation.



# AI in Medical Diagnosis

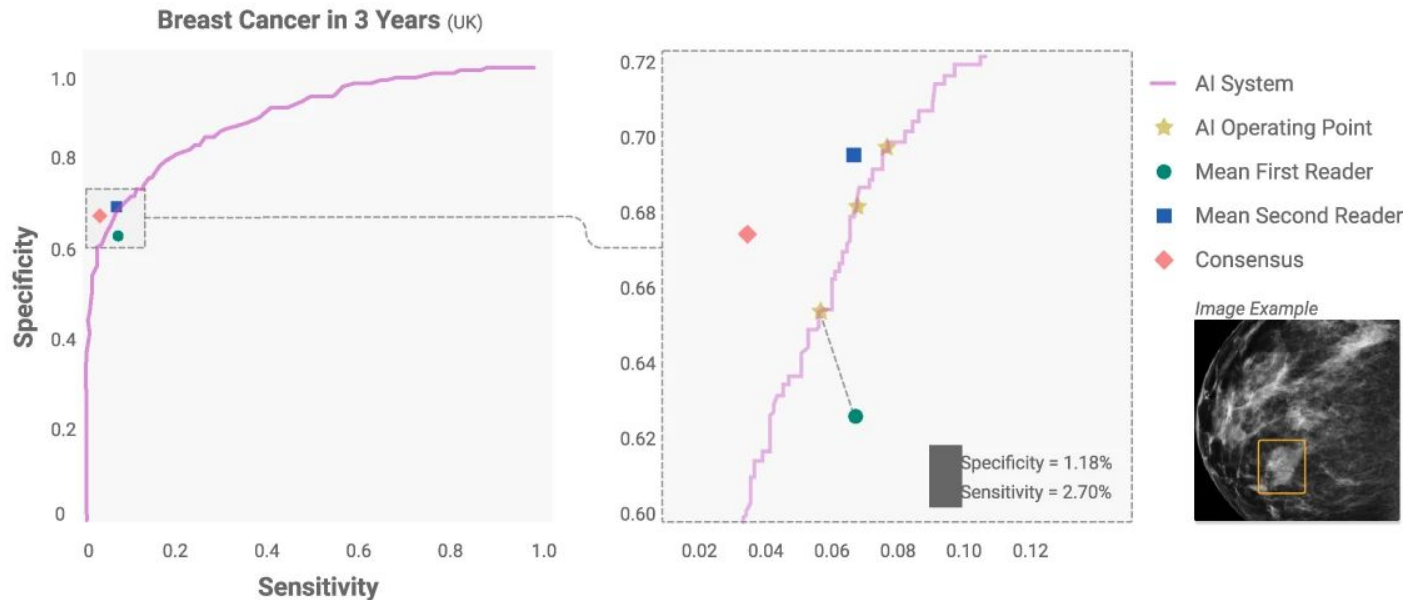
**Growing Need:** Address physician shortages, reduce medical errors, accelerate drug discovery, personalized medicine.





# AI in Medical Diagnosis

**Transformation Potential:** From diagnostics to drug discovery, deep learning is driving innovation.





# AI for Medical Diagnosis

---

**Training set** A large set of lesion images each labelled as *malignant* or *benign* (from biopsy)

**Training** Adjustment of millions parameters in *deep neural network* using the training set

**Supervised learning** For each training example, the network is told the correct label

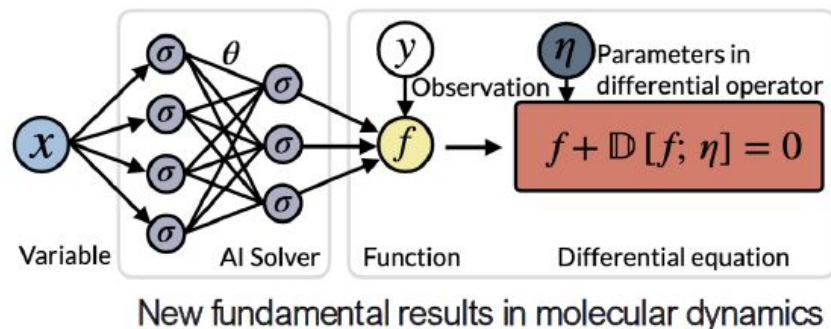
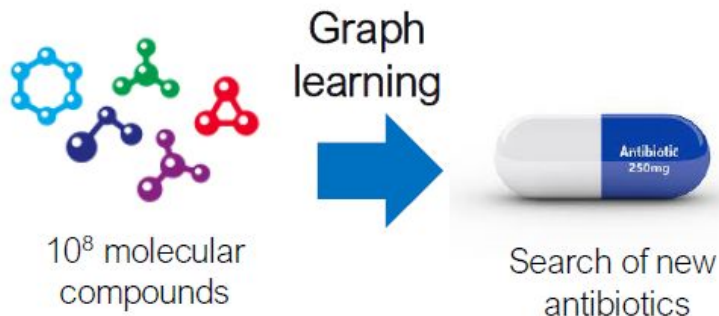
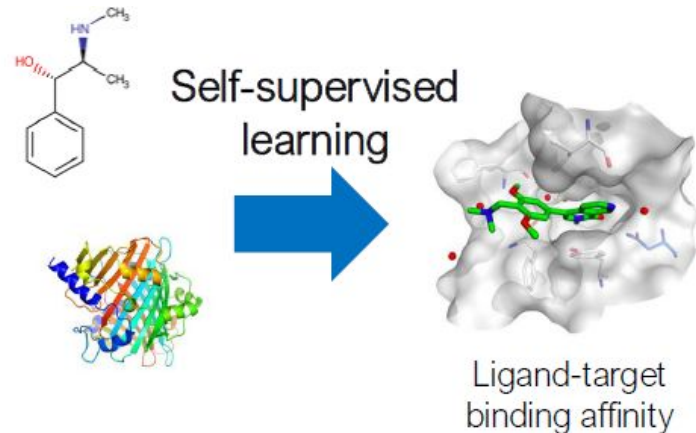
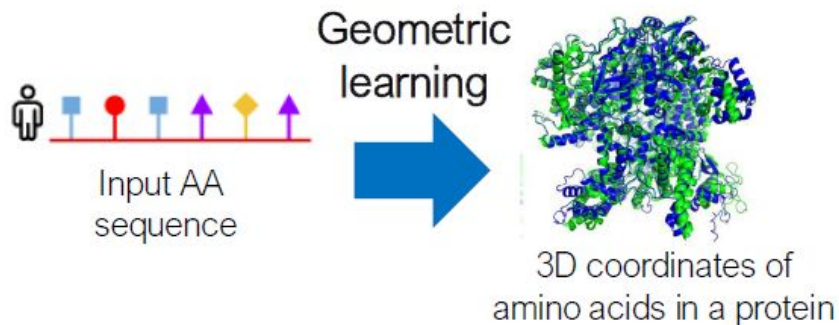
**Classification** Each input is assigned to a discrete set of classes (benign or malignant)

**Transfer learning** The deep neural network was first trained on a much larger data set of 1.28 million images of everyday objects (such as dogs, buildings, and mushrooms) and then fine-tuned on the 129,000 data set of lesion images

**Evaluation metrics** Accuracy, sensitivity, specificity, ROCAUC, confusion matrices, recall, precision, etc.

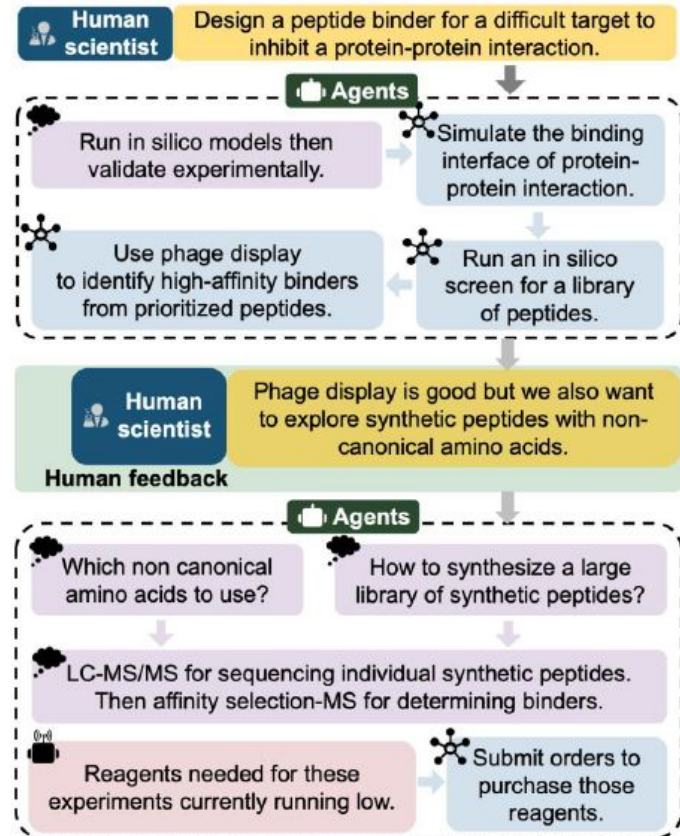
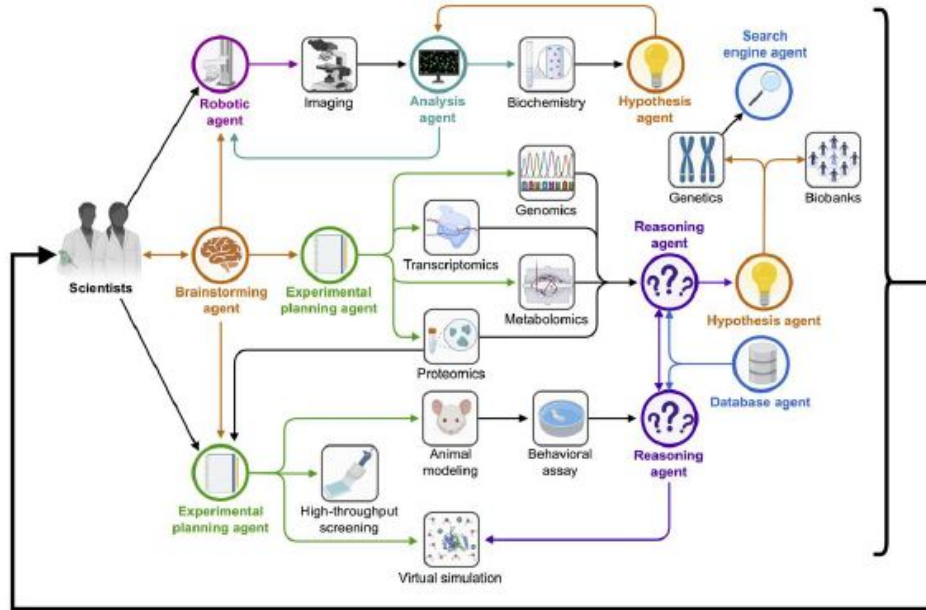
---

# AI in Medicine



# Generative AI Agents as “AI Scientist”

AI Systems can make **major scientific discoveries!**



# AI in Healthcare

Health system-scale language models are **all-purpose predictions engines**

**a**

Clinical task



Physician

## In-hospital mortality prediction

How likely is the patient to die in the hospital before discharge?

## Binned comorbidity index imputation

Without structured ICDS, how sick/chronically ill is the patient?

## 30-day all-cause readmission prediction

How likely is the patient to come back within 30 days of discharge?

Operational task



Admin

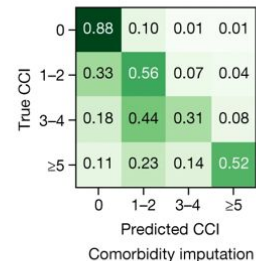
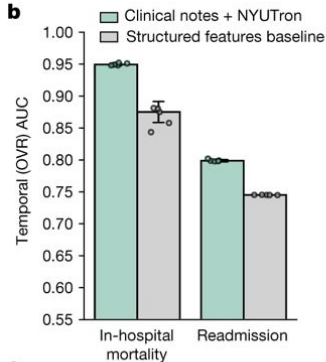
## Binned LOS prediction

How long will the patient stay in the hospital?

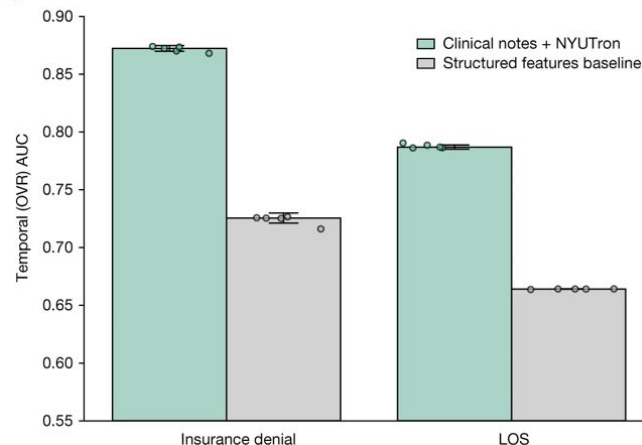
## Insurance denial prediction

How likely is the patient's insurance claim to be denied?

**b**

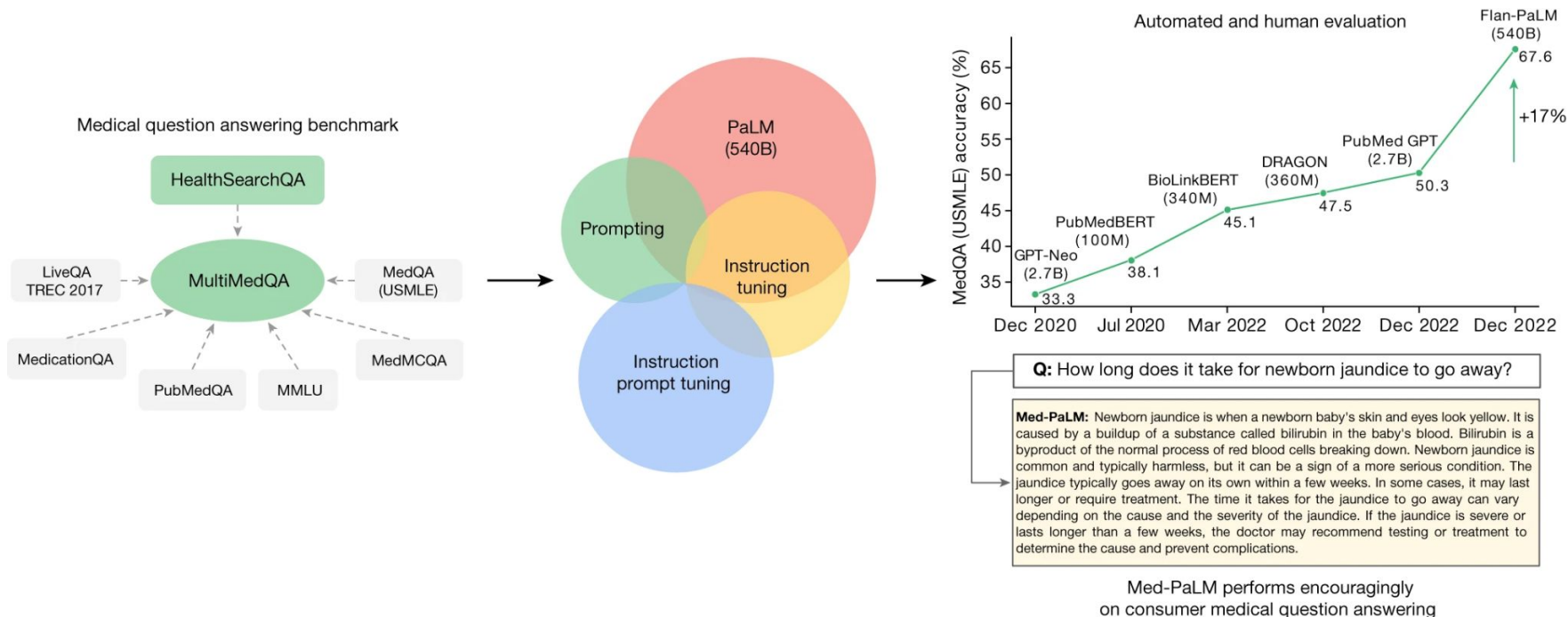


**c**



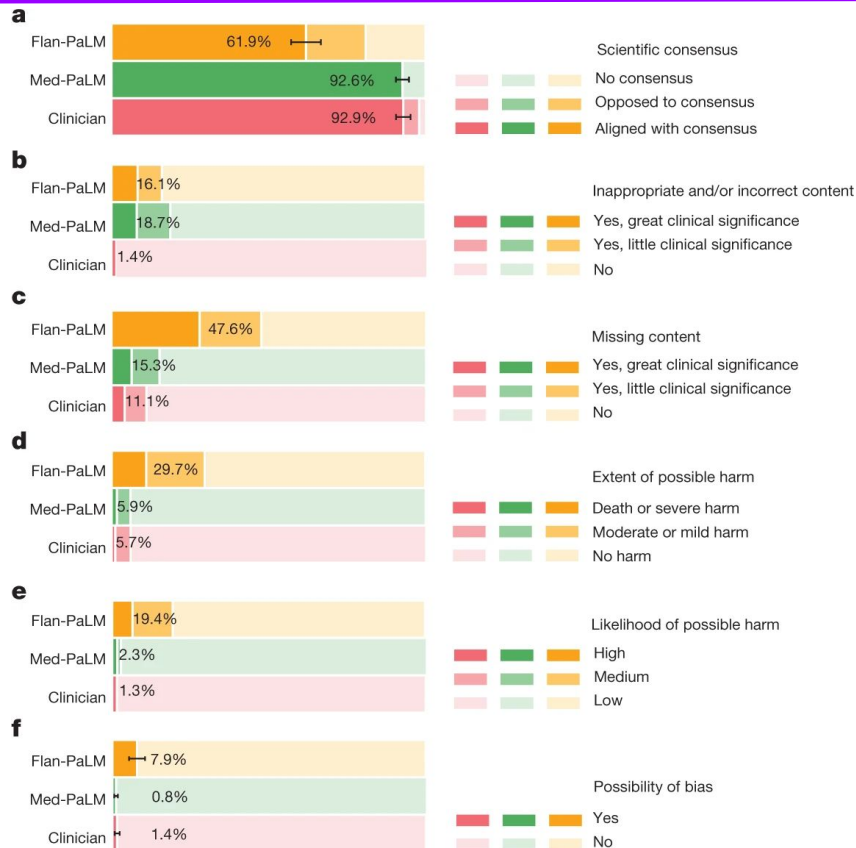
# AI in Healthcare

Large language models encode **clinical knowledge**



# AI in Healthcare

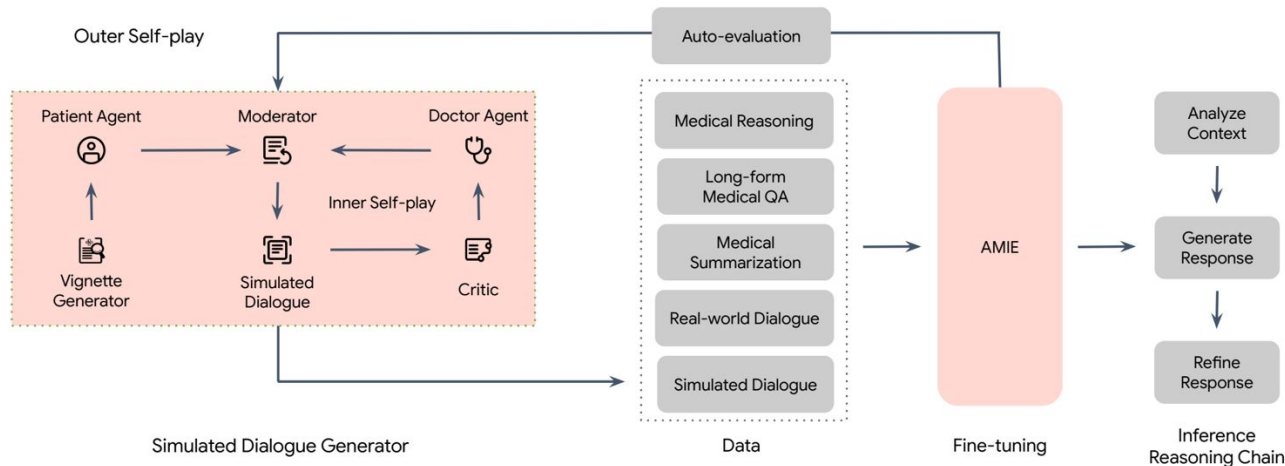
Clinicians were asked to rate answers to questions in the **HealthSearchQA**, **LiveQA** and **MedicationQA** datasets for agreement with scientific and clinical consensus.



# Conversational medical AI for diagnostic dialogue

**AMIE** is a conversational medical AI optimised for diagnostic dialogue.

AMIE is instruction fine-tuned with a combination of real-world and simulated medical dialogues, alongside a diverse set of **medical reasoning, question answering, and summarization datasets**.



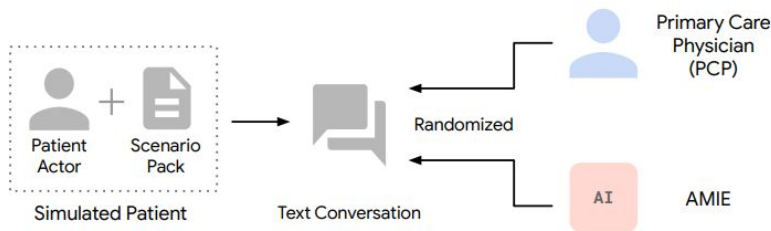
AMIE System Design



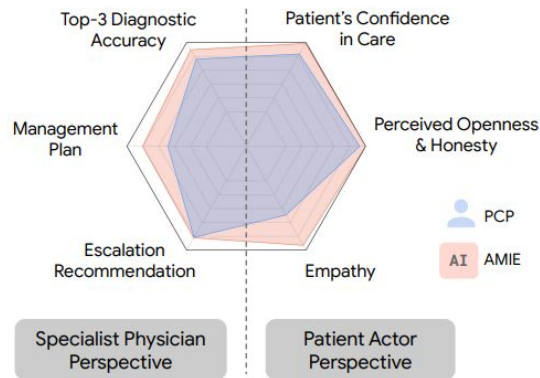
# Conversational medical AI for diagnostic dialogue

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Randomized Study Design for Remote  
Objective Structured Clinical Examination (OSCE)

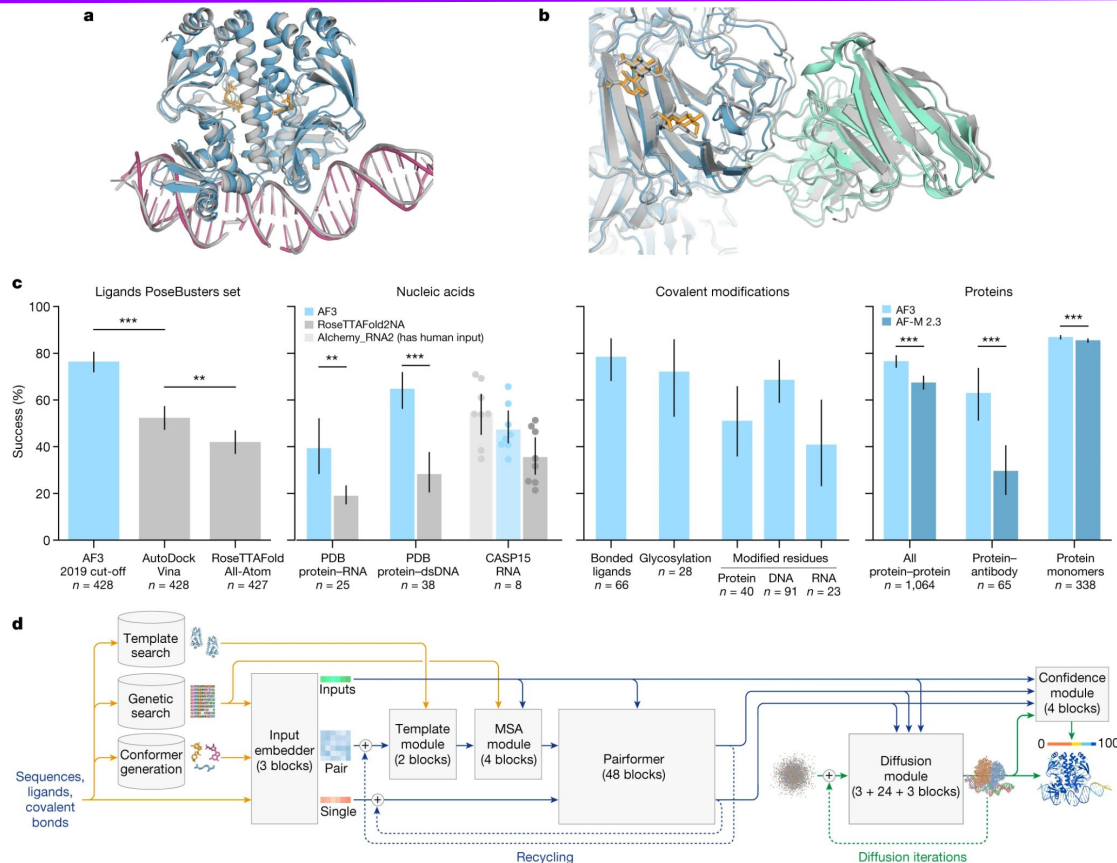


AMIE Outperforms PCPs on  
Multiple Evaluation Axes for Diagnostic Dialogue

# AI in Drug Discovery

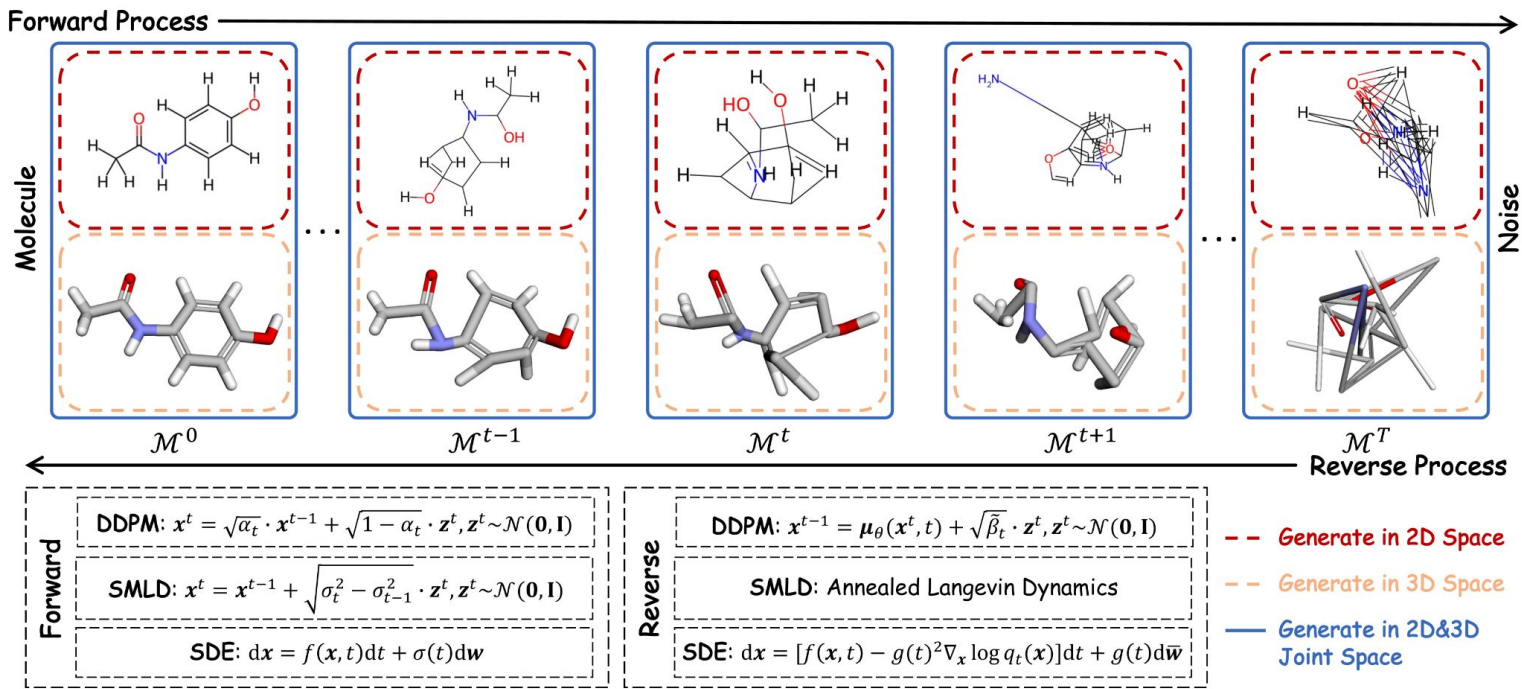
**Growing Need:** Address physician shortages, reduce medical errors, **accelerate drug discovery**, personalized medicine.

**AlphaFold 3** model with a substantially updated **diffusion-based** architecture that is capable of predicting the **joint structure of complexes** including **proteins, nucleic acids, small molecules, ions and modified residues**.



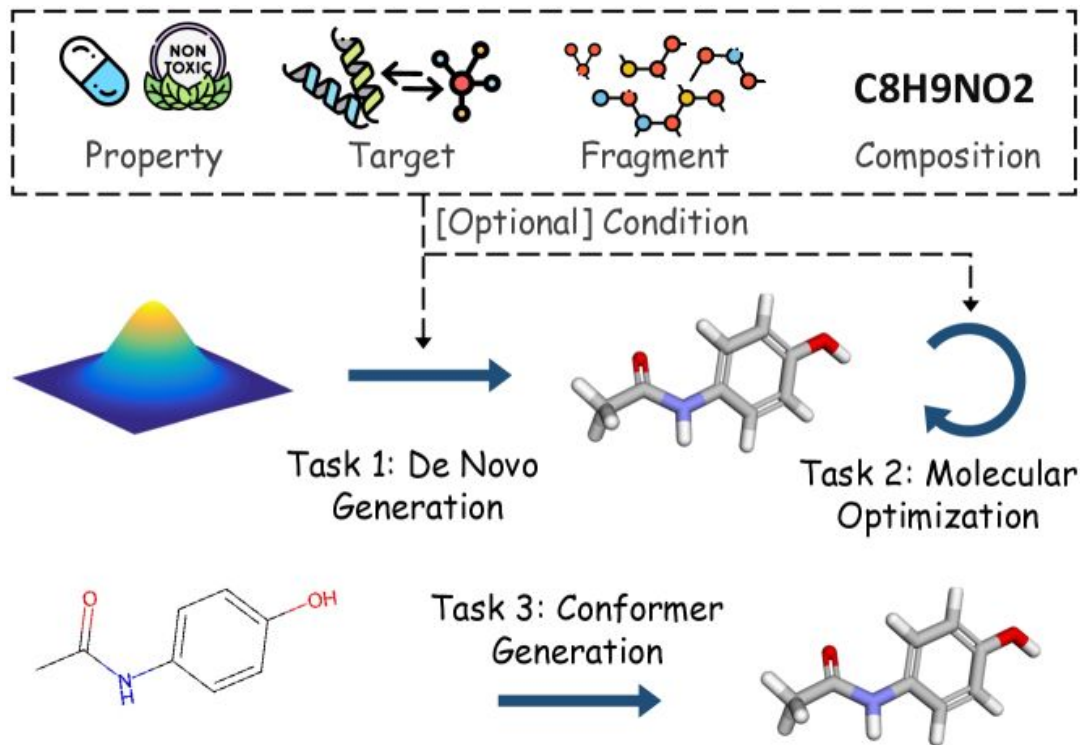
# AI in Drug Discovery

## Equivariant diffusion for small molecule generation



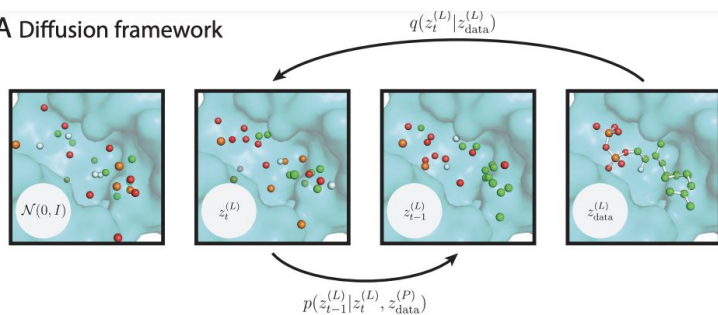
# AI in Drug Discovery

## Equivariant diffusion for small molecule generation

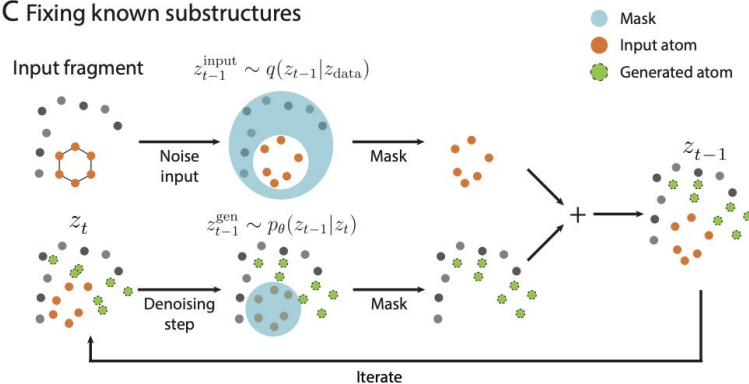


# AI in Drug Discovery

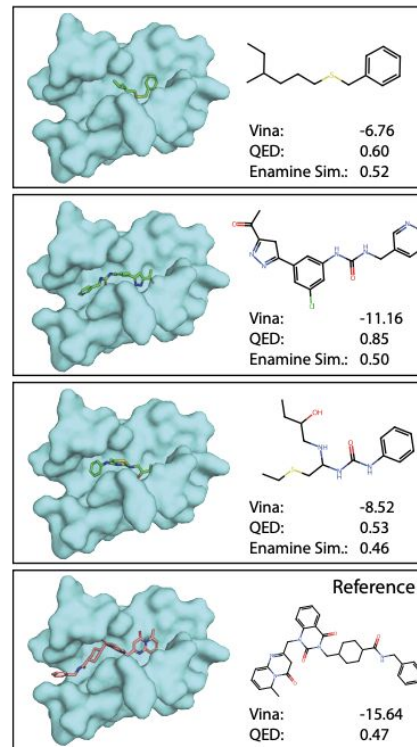
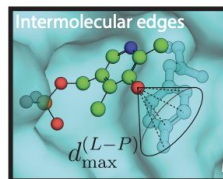
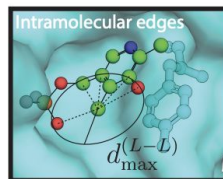
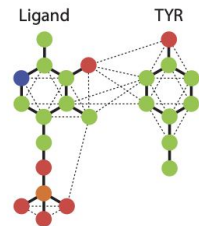
## A Diffusion framework



## C Fixing known substructures



## B Graph modeling





# AI in Drug Discovery

nature

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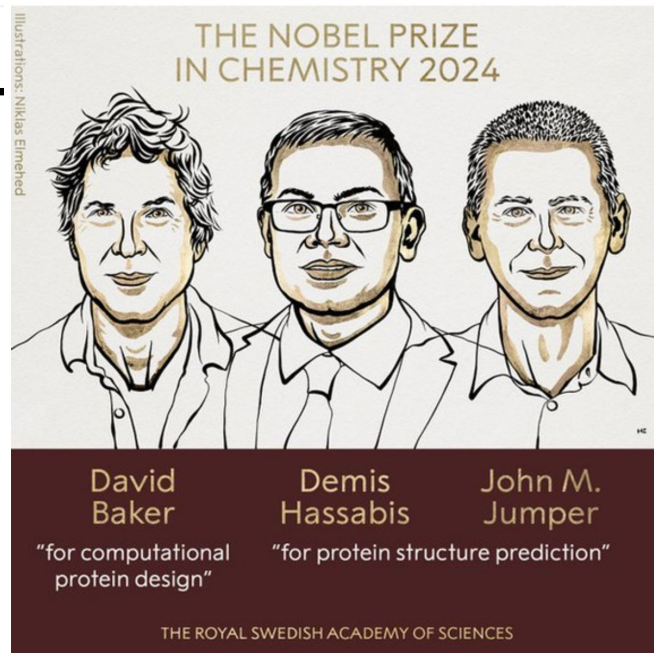
[nature](#) > [articles](#) > article

Article | [Open access](#) | Published: 11 July 2023

## De novo design of protein structure and function with RFdiffusion

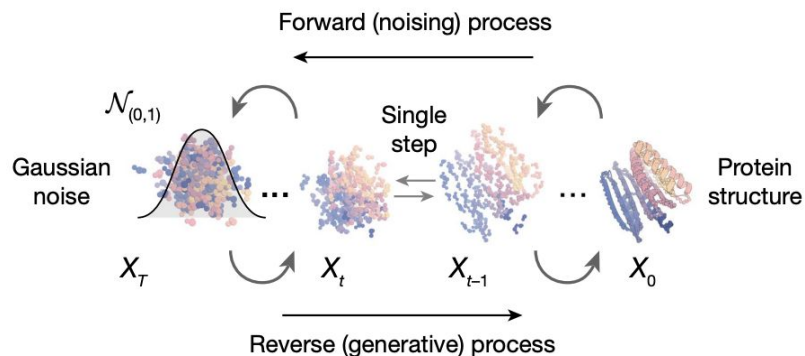
[Joseph L. Watson](#), [David Juergens](#), [Nathaniel R. Bennett](#), [Brian L. Trippe](#), [Jason Yim](#), [Helen E. Eisenach](#), [Woody Ahern](#), [Andrew J. Borst](#), [Robert J. Ragotte](#), [Lukas F. Milles](#), [Basile I. M. Wicky](#), [Nikita Hanikel](#), [Samuel J. Pellock](#), [Alexis Courbet](#), [William Sheffler](#), [Jue Wang](#), [Preetham Venkatesh](#), [Isaac Sappington](#), [Susana Vázquez Torres](#), [Anna Lauko](#), [Valentin De Bortoli](#), [Emile Mathieu](#), [Sergey Ovchinnikov](#), [Regina Barzilay](#), ... [David Baker](#)  [+ Show authors](#)

[Nature](#) **620**, 1089–1100 (2023) | [Cite this article](#)

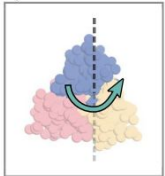


# AI in Drug Discovery

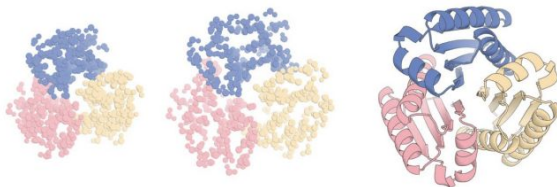
Diffusion model



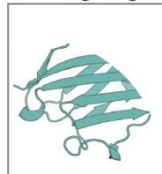
Symmetric noise



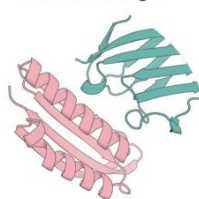
Symmetric oligomers



Binding target



Binder design



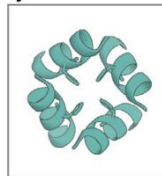
Functional motif



Motif scaffolding



Symmetric motif



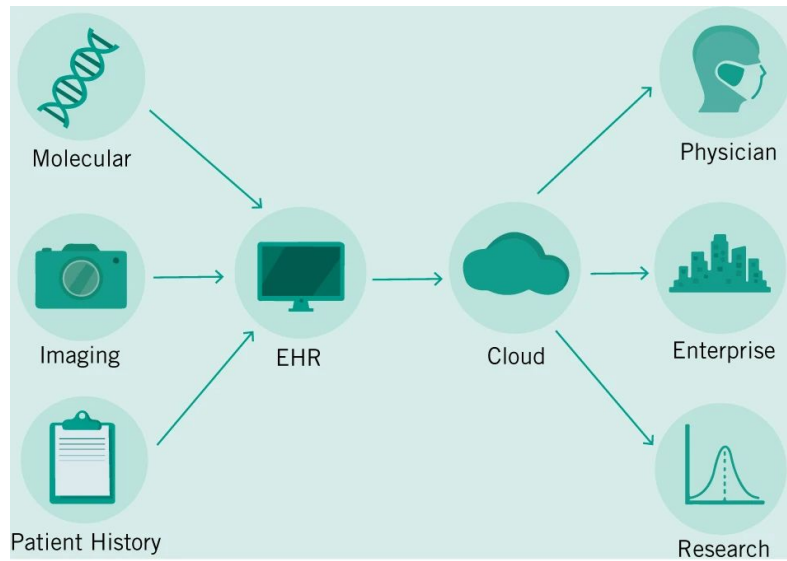
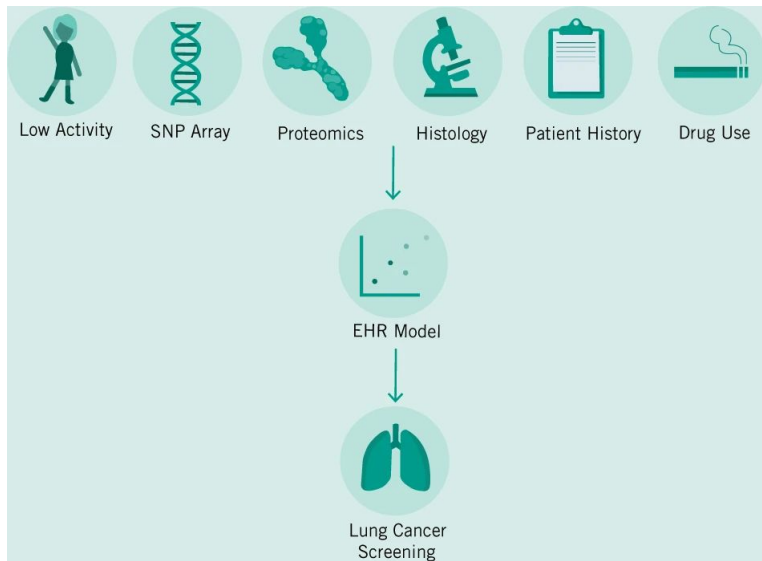
Symmetric scaffolding





# What makes biomedical data different?

**Data Availability:** Electronic Health Records (EHRs), medical imaging repositories, genomics data.



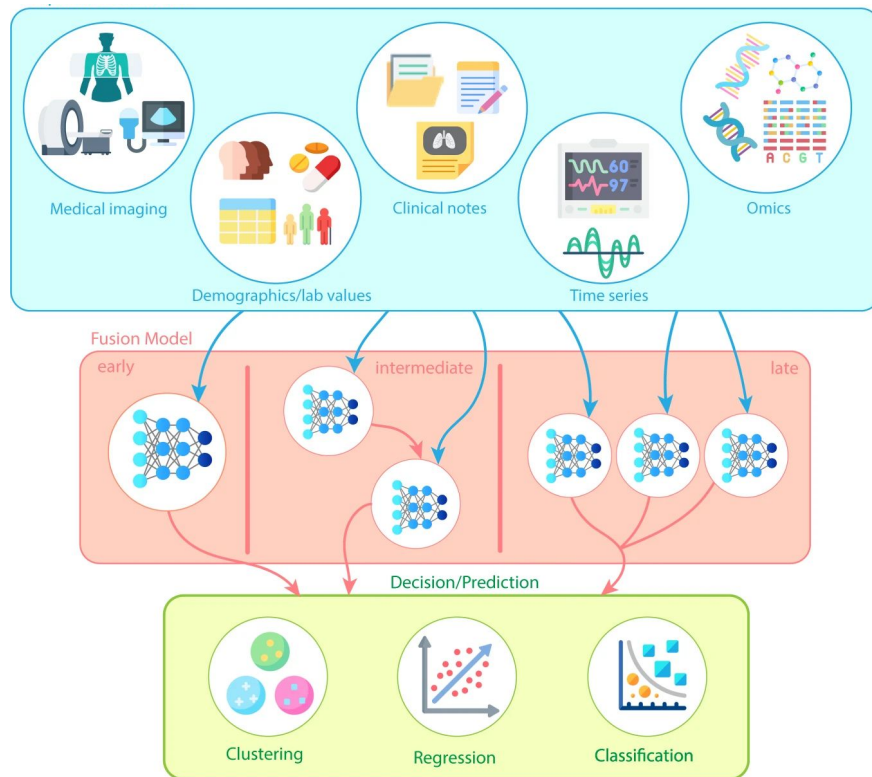
# What makes biomedical data different?

## Complexity of Healthcare Data:

Structured (EHRs, lab results), unstructured (clinical notes), image data (X-ray, MRI), multimodal data, etc.

## Accuracy alone is not sufficient

- Explainability
- Non-discriminatory predictions
- Privacy-preserving
- Causal



# What makes biomedical data different?

- Little labeled data - narrow generalization
- Lots of missing data, varying time intervals, censored labels
- Difficult to correct biases and inequities
- Motivates semi-supervised and self-supervised learning
- Motivates generative AI for rare diseases



Russakovsky et al. '14

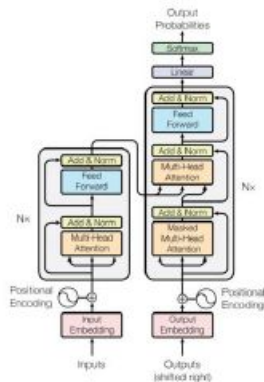


Figure 1: The Transformer - model architecture.

Vaswani et al. '18

# Medical Image Synthesis using Generative AI

---

Medical image acquisition is challenging:

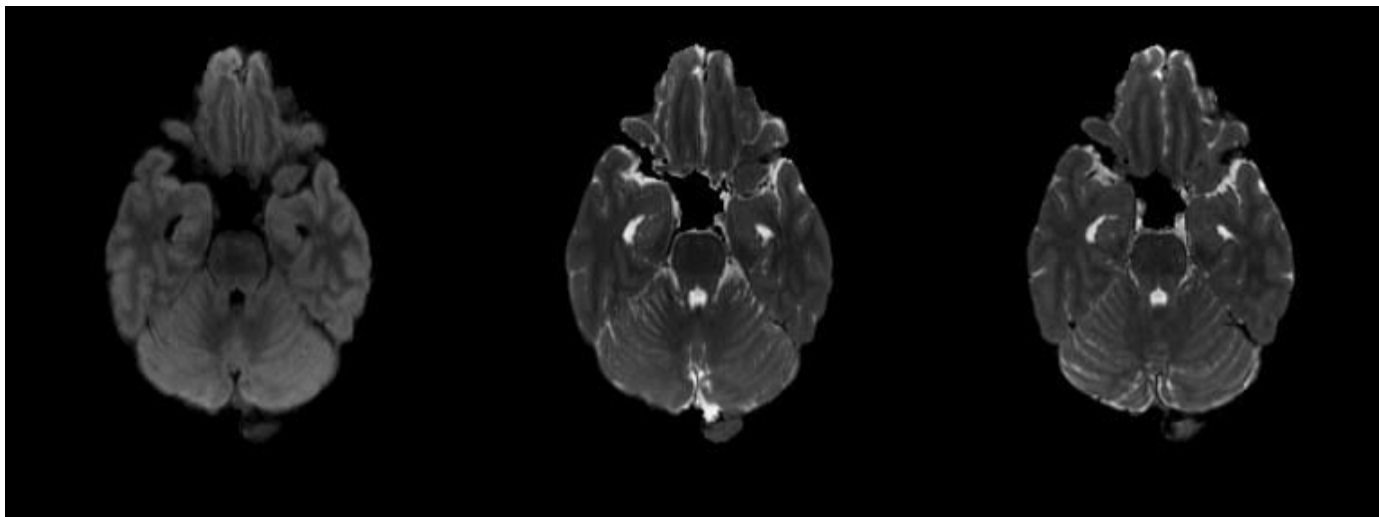
- **High operational costs** (technical fees, professional fees, facility fee)
- **High radiation exposure** (PET/CT scans expose patient to high radiation)
- **Long acquisition times** (motion artifacts due to patient movements)



# GenAI in Medical Image Synthesis

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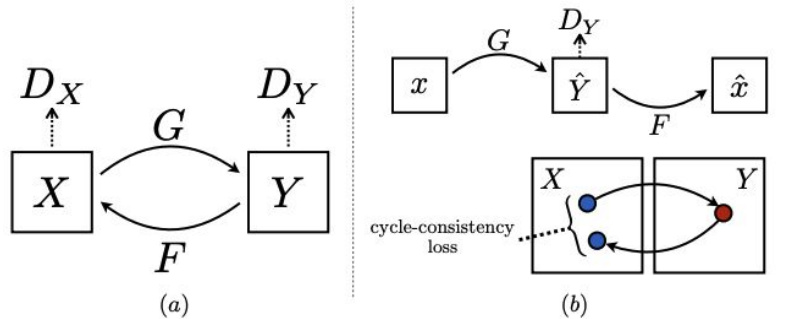
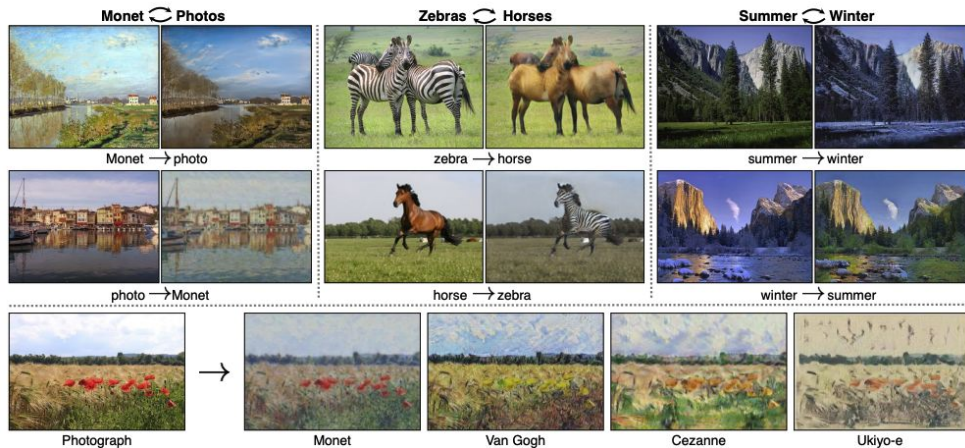
- **Low-field MRI** cuts equipment and operational costs
- **low-dose PET** reduces patient radiation exposure.
- Difficulties with **image quality**, **diagnostic accuracy**, and **practical implementation**.



# CycleGANs for unpaired image-to-image translation

## Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

Jun-Yan Zhu\*    Taesung Park\*    Phillip Isola    Alexei A. Efros  
Berkeley AI Research (BAIR) laboratory, UC Berkeley





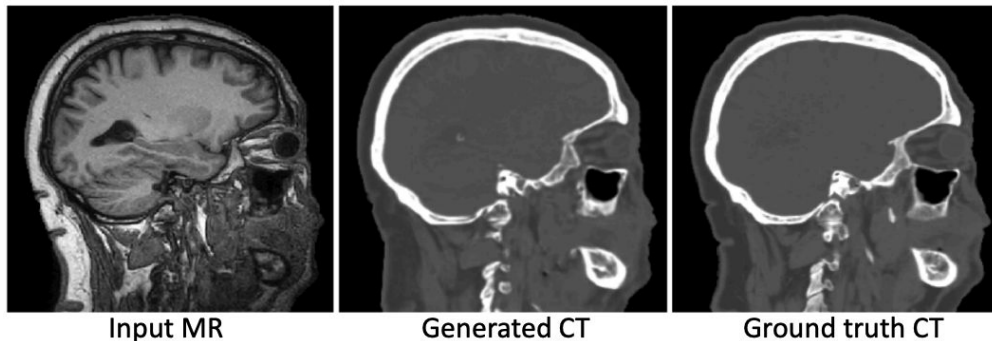
# MR to CT synthesis using CycleGANs

## Deep MR to CT Synthesis using Unpaired Data

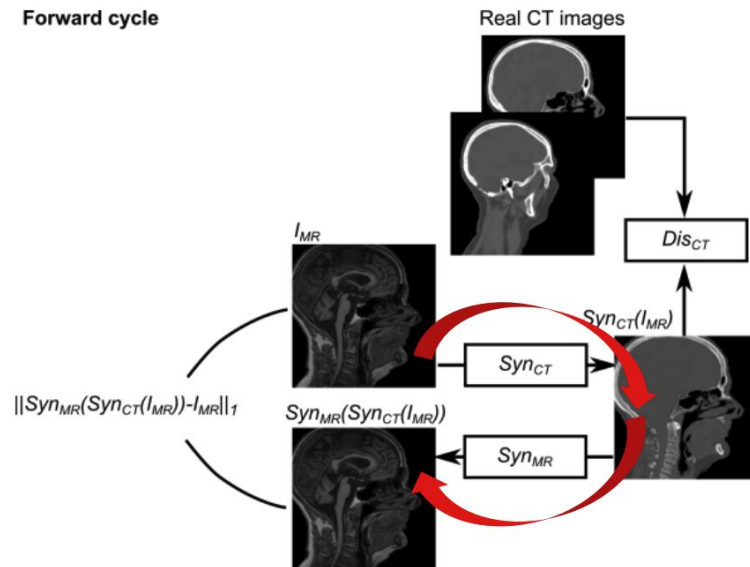
Jelmer M. Wolterink<sup>1</sup>✉, Anna M. Dinkla<sup>2</sup>, Mark H.F. Savenije<sup>2</sup>,  
Peter R. Seevinck<sup>1</sup>, Cornelis A.T. van den Berg<sup>2</sup>, Ivana Išgum<sup>1</sup>

<sup>1</sup> Image Sciences Institute, University Medical Center Utrecht, The Netherlands  
j.m.wolterink@umcutrecht.nl

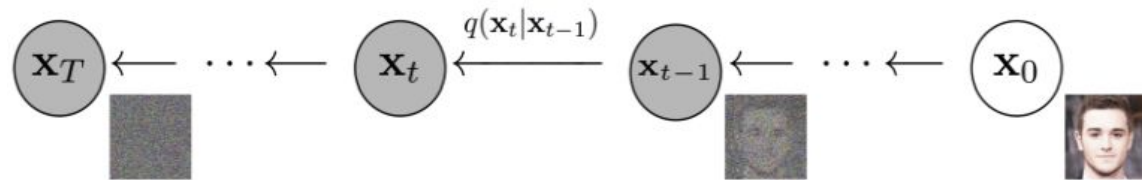
<sup>2</sup> Department of Radiotherapy, University Medical Center Utrecht, The Netherlands



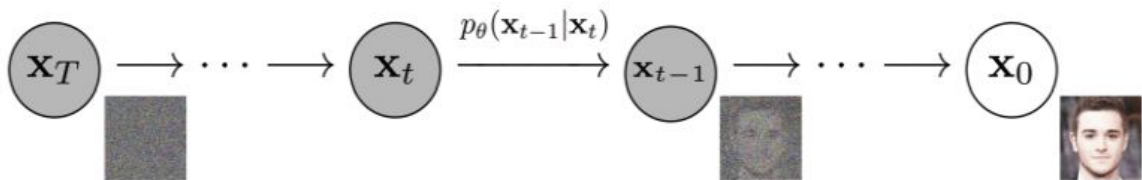
Forward cycle



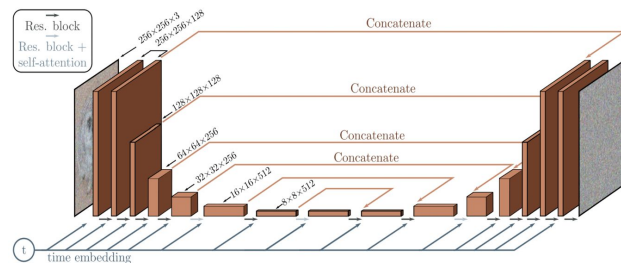
# Diffusion models for image-to-image translation



$$q(x_1, \dots, x_T | x_0) := \prod_{t=1}^T q(x_t | x_{t-1})$$
$$q(x_t | x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$



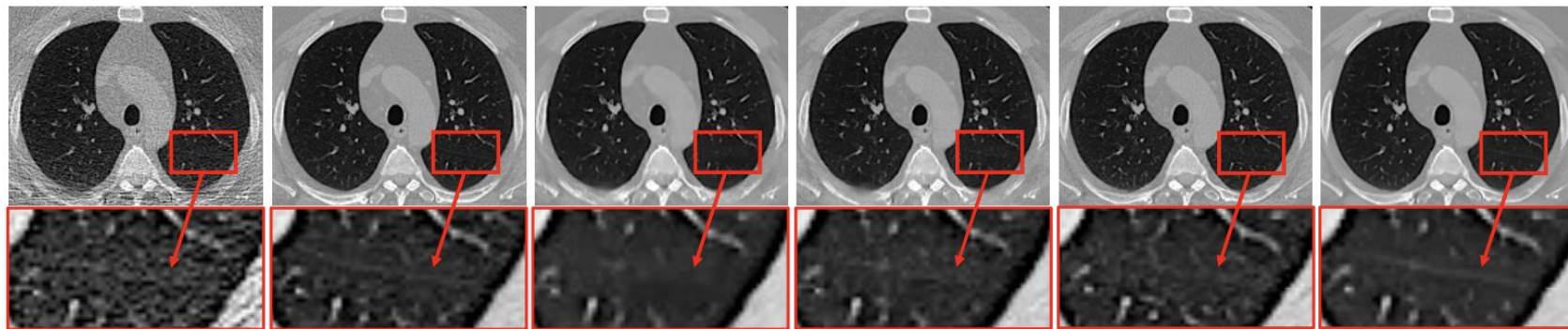
$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$



The U-Net architecture used in DDPMs

# DDPMs for medical image generation/translation

- Conversion of **low-dose (10% of normal) to full-dose lung CT**



Low-dose CT

Normal-dose CT

RED-CNN

DU-GAN

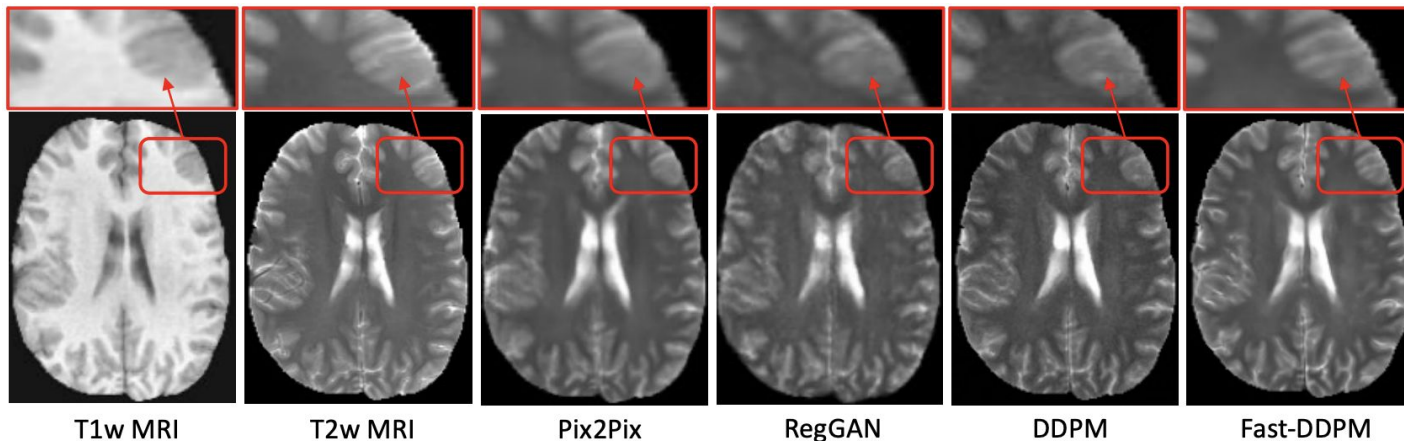
DDPM

Fast-DDPM

| Method      | PSNR        | SSIM        | Training time | Inference Time |
|-------------|-------------|-------------|---------------|----------------|
| REDCNN [2]  | 36.4        | 0.91        | 3 h           | 0.5 s          |
| DU-GAN [12] | 36.3        | 0.90        | 20 h          | 3.8 s          |
| DDPM [9]    | 35.4        | 0.87        | 141 h         | 21.4 m         |
| Fast-DDPM   | <b>37.5</b> | <b>0.92</b> | 26 h          | 12.5 s         |

# DDPMs for medical image generation/translation

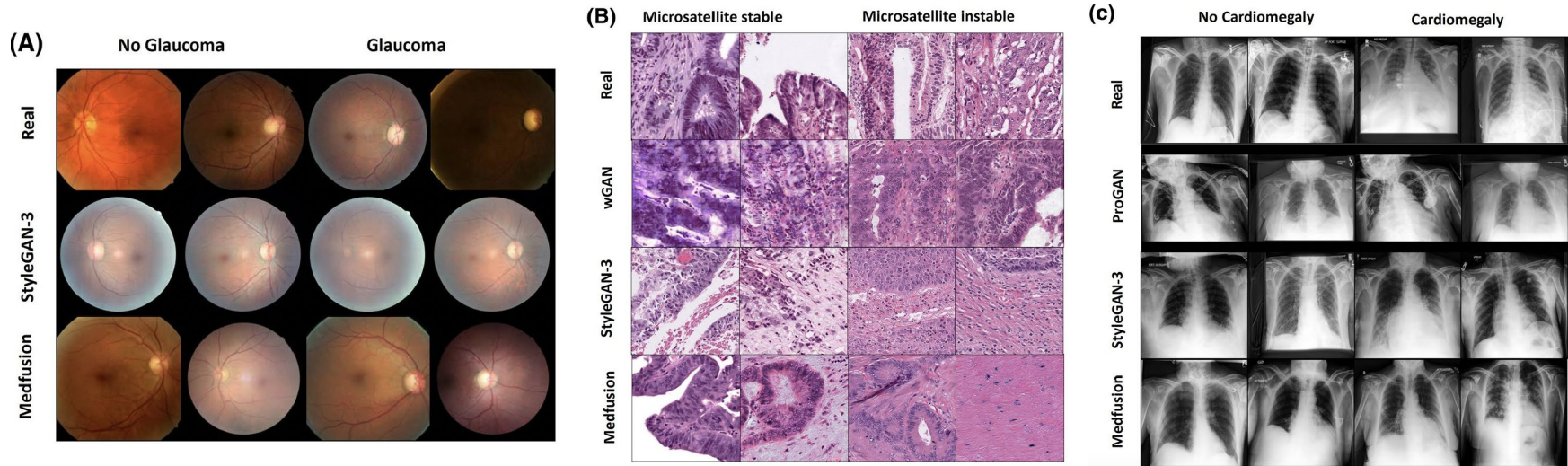
- **T1w MRI** to **T2w MRI** conversion



| Method       | PSNR        | SSIM | Training time | Inference Time |
|--------------|-------------|------|---------------|----------------|
| Pix2Pix [14] | 25.6        | 0.85 | 6 h           | 3.3 s          |
| RegGAN [19]  | 26.0        | 0.86 | 9 h           | 3.1 s          |
| DDPM [9]     | <b>26.3</b> | 0.89 | 135 h         | 22.2 m         |
| Fast-DDPM    | <b>26.3</b> | 0.89 | 27 h          | 13.2 s         |

# Latent diffusion model for medical image synthesis

- Training data simulation for **computer-aided diagnostics**

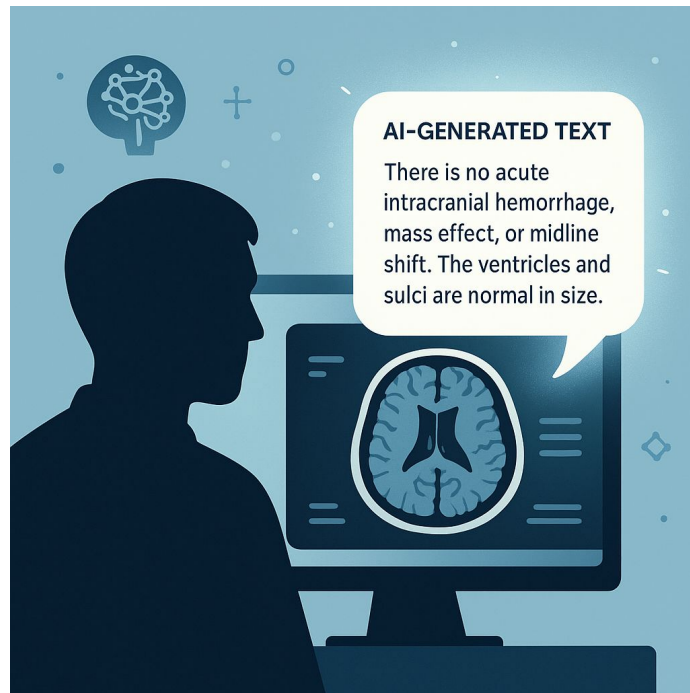




# Medical Chatbots

---

- Initialized with with general-purpose LLMs (GPT-4, LLaMA, etc.) and **fine-tuned on customized sets of biomedical instruction-following data**
  - ChatDoctor
  - Med-Alpaca
  - PMC-LLaMA
  - Clinical Camel
  - DoctorGLM
  - Huatuo
  - Visual Med-Alpaca (multimodal: text + vision)





# Medical Chatbots

---

| Question                                                                                                                                             | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the relationship between very low $Mg^{2+}$ levels, PTH levels, and $Ca^{2+}$ levels?                                                        | Very low $Mg^{2+}$ levels correspond to low PTH levels which in turn results in low $Ca^{2+}$ levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| What are the cautionary considerations and contraindications for the use of $\beta$ -blockers?                                                       | $\beta$ -blockers must be used cautiously in decompensated heart failure and are contraindicated in cardiogenic shock.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| What is the recommended next step in management for a stable child who is suspected of having intussusception and has a "target sign" on ultrasound? | The recommended next step in management for a stable child with suspected intussusception and a "target sign" on ultrasound is to undergo an ultrasound-guided air contrast enema. This procedure involves using a small amount of air and contrast material to inflate the colon and push the telescoped intestine back into place. It is a non-invasive and effective method for both diagnosing and treating intussusception in children. In some cases, the procedure may need to be repeated or followed by surgery if the intussusception cannot be reduced. It is important to treat intussusception promptly to avoid complications such as bowel obstruction, bowel perforation, or sepsis. |

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# Medical Chatbots

- United States Medical Licensing Examination (USMLE)

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## MEDALPACA - AN OPEN-SOURCE COLLECTION OF MEDICAL CONVERSATIONAL AI MODELS AND TRAINING DATA

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A PREPRINT

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Email: {michalis.papaioannou, pgrundmann, tom.oberhauser, alexei.figueroa, aloeser}@bht-berlin.de

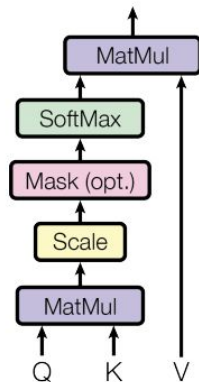
<sup>5</sup>Charité – Universitätsmedizin Berlin, corporate member of Freie Universität Berlin and Humboldt-Universität zu Berlin, Institute for Radiology, Berlin, Germany  
Email: keno-kyrill.bressem@charite.de

Table 2: Zero shot performance on the USMLE self assessment

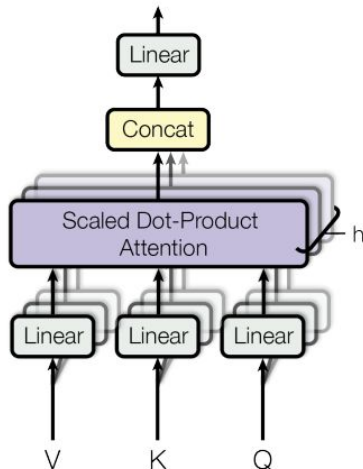
| Model                   | Step1        | Step2        | Step3        |
|-------------------------|--------------|--------------|--------------|
| LLaMA 7b [15]           | 0.198        | 0.202        | 0.203        |
| Alpaca 7b naive [11]    | 0.275        | 0.266        | 0.293        |
| Alpaca 7b LoRA          | 0.220        | 0.138        | 0.252        |
| MedAlpaca 7b            | 0.297        | 0.312        | 0.398        |
| MedAlpaca 7b LoRA       | 0.231        | 0.202        | 0.179        |
| MedAlpaca 7b LoRA 8bit  | 0.231        | 0.241        | 0.211        |
| ChatDoctor (7b) [10]    | 0.187        | 0.185        | 0.148        |
| LLaMA 13b [15]          | 0.222        | 0.248        | 0.276        |
| Alpaca 13b naive        | 0.319        | 0.312        | 0.301        |
| MedAlpaca 13b           | <b>0.473</b> | <b>0.477</b> | <b>0.602</b> |
| MedAlpaca 13b LoRA      | 0.250        | 0.255        | 0.255        |
| MedAlpaca 13b LoRA 8bit | 0.189        | 0.303        | 0.289        |

# Attention is all you need!

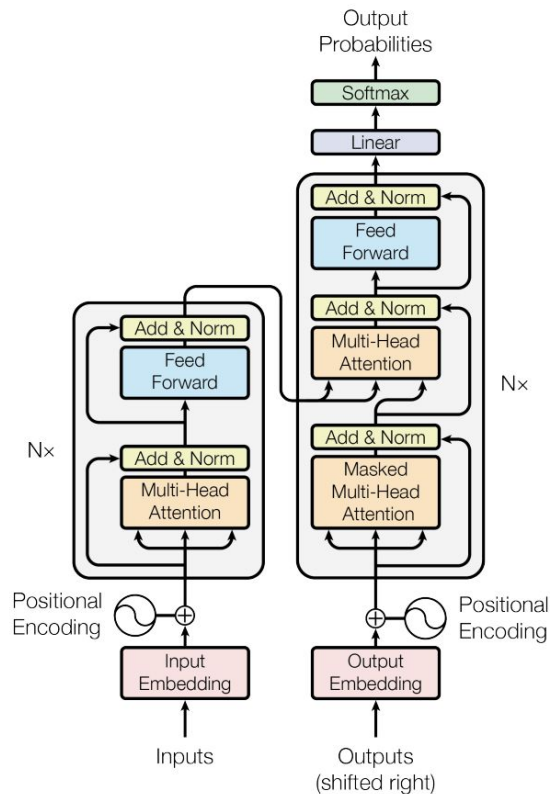
Scaled Dot-Product Attention



Multi-Head Attention

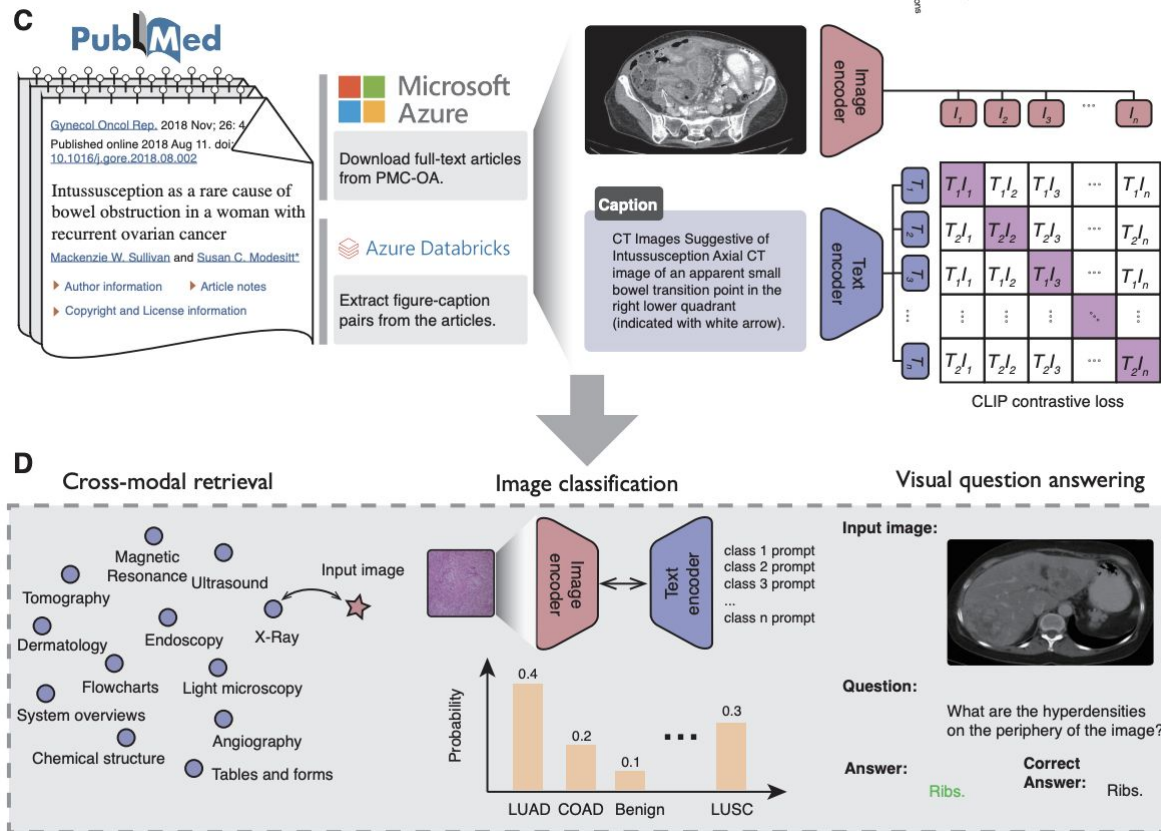


$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad \sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

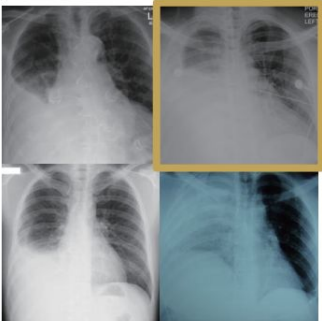
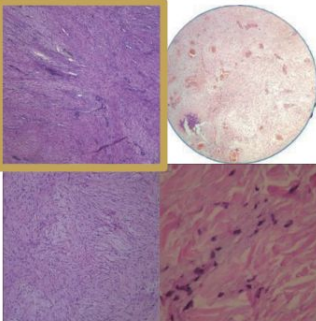
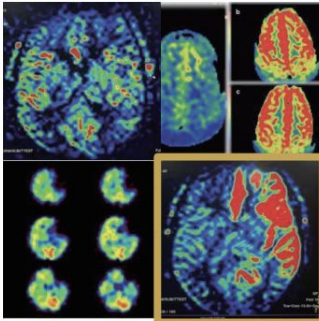
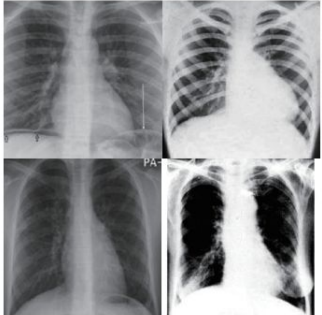
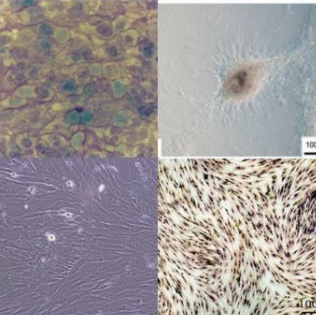
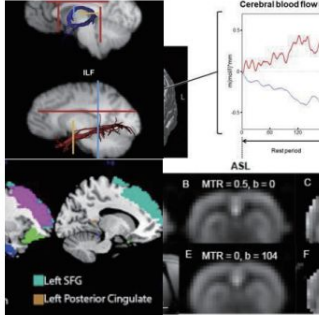


# Multimodal biomedical foundation models

- BiomedCLIP



# Medical Image Captioning

|            |                                                                                                      |                                                                                                                                                                             |                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|            | <p><b>Caption</b></p> <p>Chest X-ray on admission showing a large pleural effusion on the right.</p> | <p><b>Caption</b></p> <p>Photomicrograph showing proliferation of spindle cells with wavy nuclei in collagenous to myxoid stroma (Hematoxylin and Eosin staining, 40x).</p> | <p><b>Caption</b></p> <p>ASL image shows increased cerebral blood flow (CBF) in left fronto-parieto-temporal region.</p> |
| BiomedCLIP |                     |                                                                                           |                                       |
| CLIP       |                    |                                                                                          |                                      |



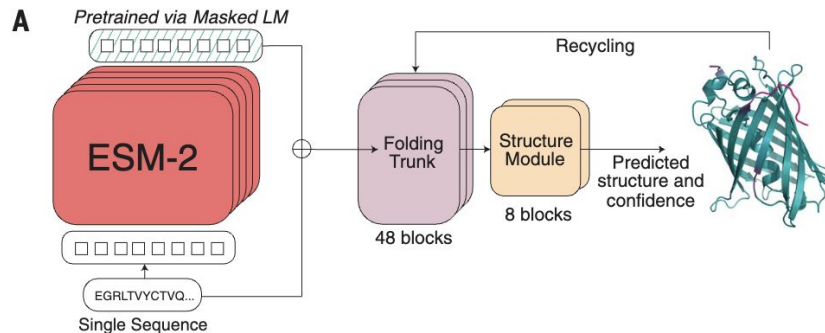
# Protein language models (PLM)

## STRUCTURE PREDICTION

### Evolutionary-scale prediction of atomic-level protein structure with a language model

Zeming Lin<sup>1,2†</sup>, Halil Akin<sup>1†</sup>, Roshan Rao<sup>1†</sup>, Brian Hie<sup>1,3†</sup>, Zhongkai Zhu<sup>1</sup>, Wenting Lu<sup>1</sup>, Nikita Smetanin<sup>1</sup>, Robert Verkuil<sup>1</sup>, Ori Kabeli<sup>1</sup>, Yaniv Shmueli<sup>1</sup>, Allan dos Santos Costa<sup>4</sup>, Maryam Fazel-Zarandi<sup>1</sup>, Tom Sercu<sup>1</sup>, Salvatore Candido<sup>1</sup>, Alexander Rives<sup>1,2\*</sup>

Recent advances in machine learning have leveraged evolutionary information in multiple sequence alignments to predict protein structure. We demonstrate direct inference of full atomic-level protein structure from primary sequence using a large language model. As language models of protein sequences are scaled up to 15 billion parameters, an atomic-resolution picture of protein structure emerges in the learned representations. This results in an order-of-magnitude acceleration of high-resolution structure prediction, which enables large-scale structural characterization of metagenomic proteins. We apply this capability to construct the ESM Metagenomic Atlas by predicting structures for >617 million metagenomic protein sequences, including >225 million that are predicted with high confidence, which gives a view into the vast breadth and diversity of natural proteins.





# AI Agents propose strategy for antibody design

## (a) Phase 1: Team selection



Human  
Researcher



Principal  
Investigator



Immunologist



Machine  
Learning  
Specialist



Computational  
Biologist



Scientific  
Critic

## (b) Phase 2: Project specification

### Team Meeting



### Summary

Modify nanobodies Ty1, H11-D4, Nb21, and VHH-72 to improve binding to KP.3.

## (c) Phase 3: Tools selection

### Team Meeting



### Summary

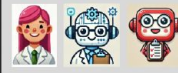
ESM, AlphaFold-Multimer, and Rosetta to design improved nanobodies.

## (d) Phase 4: Tools implementation

### ESM

QVQLVE... → DVQLVE...  
ESM LLR = 3.66

### Individual Meeting



### Summary

Python script to compute ESM log-likelihood ratios for single point mutations.

### AlphaFold-Multimer

DVQLVE... →   
AF ipLDDT = 76.52

### Individual Meeting



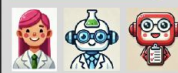
### Summary

Python script to extract AlphaFold-Multimer interface pLDDT.

### Rosetta

→   
RS dG = -37.91

### Individual Meeting



### Summary

Python and XML scripts to compute binding energies with Rosetta.

# Conclusions

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- **Enhanced Diagnostic Accuracy:** Deep learning models excel in medical imaging, detecting anomalies with precision, often surpassing human performance.
  - **Accelerated Drug Discovery:** Generative AI speeds up the design of novel drugs, **reducing time and cost** in the development pipeline.
  - **Personalized Medicine:** AI enables **tailored treatments** by predicting patient-specific responses, improving outcomes, and minimizing side effects.
  - **Scalable and Equitable Healthcare:** AI-powered solutions improve access to care in under-resourced areas through **telemedicine** and **predictive analytics**.
  - **Data-Driven Insights:** Unlocks patterns in large-scale health data, **supporting evidence-based decisions** and advancing medical research.
-